

WASP-2b

R-0163

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_170608 · created 2026-07-16T16:02:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457912.8712 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.045
Transit depth (Rp/Rs)^2	1.32 +/- 1.05 [%]
Orbital Inclination (inc)	88.81 +/- 2.94
Airmass coefficient 1 (a1)	0.94 +/- 0.52
Airmass coefficient 2 (a2)	0.02 +/- 0.44
Scatter in the residuals of the lightcurve fit is	52.95 %
Best Comparison Star	#3 - [491, 290]
Optimal Aperture	9.41
Optimal Annulus	19.6
Transit Duration (day)	0.073 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.115 $\pm$ 0.045 vs 0.1326 (deviation 0.39 σ)
Duration fit vs published	0.073 d vs 0.0737 d (deviation 0.06 σ)
Mid-time O-C	10.6 min
Depth SNR	2.6
Residual scatter	52.95 %

Light curve

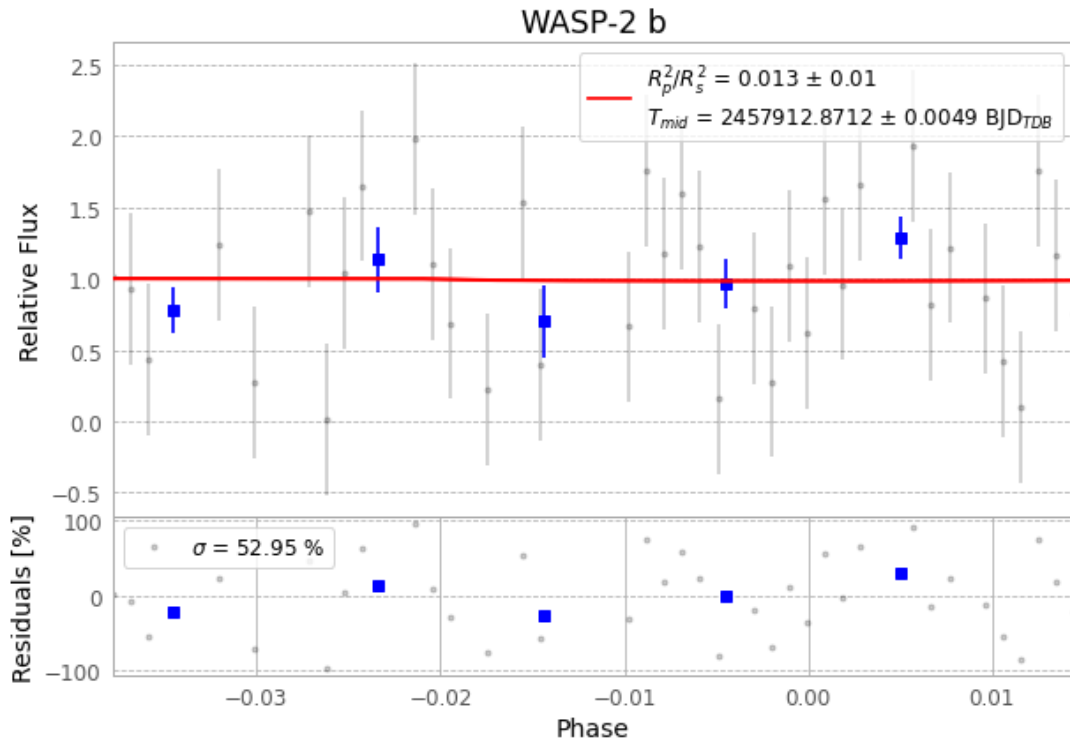


Figure 1 — FinalLightCurve_WASP-2 b_2017-06-07.png (SHA-256 3952e3d651e91d75...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [33, 24], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5ee2b477b030b192...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1634555

Data provenance

90 FITS frames (59 MB), WASP-2170608064813.FITS → WASP-2170608111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-170608.log (b1956c1b94bd...), FinalLightCurve_WASP-2 b_2017-06-07.png (3952e3d651e9...), FinalLightCurve_WASP-2 b_2017-06-07.csv (997aad6b0c60...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 16:02Z.